

## Review

$X \sim \mathcal{P}(\mu)$  : Poisson distribution if  $P\{X=n\} = \pi_n(\mu) = \frac{e^{-\mu} \mu^n}{n!}$ .

$$E(X) = \text{Var}(X) = \mu.$$

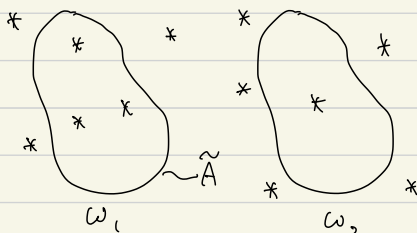
$\Pi: \Omega \rightarrow S^\infty$  random countable subset

$$A \subseteq S \quad N(A): \Omega \longrightarrow \{0, 1, 2, \dots, \infty\} \quad N(A) = \#\{\Pi \cap A\}$$

$$\omega \longmapsto \#\{\Pi(\omega) \cap A\}$$

$$\omega_1: \text{---} \underset{0}{\bullet} \text{---} [ \text{---} * * * ]_1 \text{---} * * \text{---} \quad A = [0, 1] \quad N(A)(\omega_1) = 2$$

$$\omega_2: \text{---} \underset{0}{\bullet} \text{---} [ \text{---} ]_1 \text{---} * * * \text{---} \quad N(A)(\omega_2) = 0$$



$$N(\tilde{A})(\omega_1) = 3$$

$$N(\tilde{A})(\omega_2) = 1$$

## Measurability ( $\mathcal{R}, \mathcal{F}, \mathbb{P}$ )

$\mathcal{F}$  is a  $\sigma$ -field ( $\sigma$ -algebra) if ①  $\emptyset \in \mathcal{F}$  ②  $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

$$\text{③ } A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

If  $A \in \mathcal{F}$ , we say  $A$  is a measurable (m'ble) set.

$A$  is m'ble

$$\text{Prop ① } \Omega \in \mathcal{F} \quad \text{② } A_1, \dots, A_n \in \mathcal{F} \Rightarrow \bigcup_{i=1}^n A_i \in \mathcal{F} \quad \text{③ } A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

$$\text{Ex } \Omega = \mathbb{R}^d, \mathcal{B}(\mathbb{R}^d) = \text{Borel } \sigma\text{-algebra}$$

= smallest  $\sigma$ -algebra which contains all open sets.

$$A \in \mathcal{B}(\mathbb{R}^d) \Rightarrow A: \text{Borel set.}$$

Q) What does a Borel set look like?

$$(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2) \Rightarrow \mathcal{F}_1 \times \mathcal{F}_2: \text{smallest } \sigma\text{-algebra which contains } \forall A \times B, \\ A \in \mathcal{F}_1, B \in \mathcal{F}_2$$

Thm  $E \in \mathcal{F}_1 \times \mathcal{F}_2 \Rightarrow E_x \in \mathcal{F}_2$  and  $E^y \in \mathcal{F}_1$   $\forall x \in \Omega_1, y \in \Omega_2$ .

where  $E_x = \{y : (x, y) \in E\}$ ,  $E^y = \{x : (x, y) \in E\}$ .

pf)  $\mathcal{A} = \{E \in \mathcal{F}_1 \times \mathcal{F}_2 : E_x \in \mathcal{F}_2\}$ .

Check ①  $A \times B \in \mathcal{A}$  for  $\forall A \in \mathcal{F}_1, B \in \mathcal{F}_2$

②  $\mathcal{A}$  is  $\sigma$ -algebra.

$\Rightarrow \mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{A} \quad \therefore \forall E \in \mathcal{F}_1 \times \mathcal{F}_2, E_x \in \mathcal{F}_2$

Def A fn  $f: \Omega_1 \rightarrow \Omega_2$  is m'ble if  $f^{-1}(B)$  is m'ble for  $\forall$  m'ble  $B \subseteq Y$ .

Now, fix  $(\Omega, \mathcal{F}, \mathbb{P})$ ,  $(S, \Sigma)$  <sup>state space</sup>

We assume  $D = \{(x, y) : x = y\} \subseteq S \times S$  is m'ble  <sup>$\Sigma$   $\sigma$ -alg.</sup>

$\Rightarrow D_x = \{x\}$  is m'ble for  $\forall x \in S$ .

In this chapter,  $S = \mathbb{R}^d$ ,  $\Sigma = \mathcal{B}(\mathbb{R}^d)$  (Chapter 1,  $S$ : manifold)

## II Poisson Process

$\Pi: \Omega \rightarrow S^\infty$  is a Poisson process if

(i) for any disjoint m'ble sets  $A_1, A_2, \dots, A_n$  of  $S$ , the random variables  $N(A_1), N(A_2), \dots, N(A_n)$  are independent.

(ii)  $N(A)$  has Poisson distribution  $\mathcal{P}(\mu)$  where  $\mu = \mu(A)$ ,  $0 \leq \mu \leq \infty$ .

$\mu(A) = \mathbb{E}[N(A)]$ .

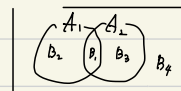
If  $A_1, A_2, \dots$  disjoint,  $A = \bigcup_{i=1}^{\infty} A_i$ , then  $N(A) = \sum_{n=1}^{\infty} N(A_n)$

Take  $\mathbb{E}$  on both sides  $\Rightarrow \mu(A) = \sum_{n=1}^{\infty} \mu(A_n)$ .  $\mu(\emptyset) = 0$ .

$\mu$ : "mean measure" of  $\Pi$ .

Ex) How to calculate  $\text{Cov}(N(A_1), N(A_2))$ ?

$$\text{Cov}(N(A_1), N(A_2)) = \mathbb{E}[N(A_1)N(A_2)] - \underbrace{\mathbb{E}[N(A_1)]}_{=\mu(A_1)} \underbrace{\mathbb{E}[N(A_2)]}_{=\mu(A_2)}$$



$$\begin{aligned} \mathbb{E}[N(A_1)N(A_2)] &= \mathbb{E}[(N(B_1) + N(B_2))(N(B_1) + N(B_3))] \\ &= \mathbb{E}[N(B_1)^2 + N(B_1)N(B_3) + N(B_2)N(B_1) + N(B_2)N(B_3)] \end{aligned}$$

$$\begin{aligned}
 &= \mu(B_1)^2 + \mu(B_1) + \mu(B_1)\mu(B_3) + \mu(B_1)\mu(B_2) + \mu(B_2)\mu(B_3) \\
 &= (\mu(B_1) + \mu(B_2))(\mu(B_1) + \mu(B_3)) + \mu(B_1) \\
 &= \mu(A_1)\mu(A_2) + \mu(A_1 \cap A_2)
 \end{aligned}$$

$$\therefore \text{Cov}(N(A_1), N(A_2)) = \mu(A_1 \cap A_2)$$

\* Property of mean measure

Suppose that  $\mu(\{x\}) = m > 0 \Rightarrow P(N(\{x\}) \geq 2) = [-e^{-m} - me^{-m}] > 0$  ( $\neg$  X)

$\therefore \mu(\{x\}) = 0$  for  $\forall x \in S$  : "non-atomic".

(Q)  $\mu$  is non-atomic  $\Rightarrow \mu$  is mean measure (Section 2.5)

(Ex)  $\mu(A) = \int_A \lambda(x) dx$  : non-atomic.  $\lambda$  = rate, intensity  
 $\uparrow$  positive m'ble f'n

If  $\lambda(x) = \lambda$ ,  $\mu(A) = \lambda \text{Vol}(A)$  : uniform or homogeneous Poisson process

(Ex)  $S = \mathbb{R}$   $\mu(t) = \begin{cases} \mu(\cdot, t] = \mathbb{E}[N(0, t)] & (t \geq 0) \\ -\mu(t, \cdot] = -\mathbb{E}[N(t, 0)] & (t < 0) \end{cases}$

$$\mu(a, b] = \mu(b) - \mu(a)$$

$\uparrow$   
 Stieltjes measure

$\mu$  should be continuous ( $\because$  non-atomic)

## [2] Superposition Thm

Thm Let  $\Pi_1, \Pi_2, \dots$  be a countable collection of independent Poisson processes on  $S$  and let  $\Pi_n$  have mean measure  $\mu_n$  for each  $n$ .

Then  $\Pi = \bigcup_{n=1}^{\infty} \Pi_n$  is a Poisson process with mean measure  $\mu = \sum_{n=1}^{\infty} \mu_n$

## Disjointness Lemma

$\Pi_1, \Pi_2$  : independent Poisson process on  $S$ ,  $A$  : m'ble,  $\mu_1(A), \mu_2(A) < \infty$   
 $\Rightarrow \Pi_1$  and  $\Pi_2$  are disjoint with probability 1 on  $A$ :

$$P(\Pi_1 \cap \Pi_2 \cap A = \emptyset) = 1.$$

Sketch of proof

$A^\dagger$ : set of all finite subsets  $\Lambda$  of  $A$ .  $\Pi_1 \cap A : p_1$   $\Pi_2 \cap A : p_2$

Define  $\eta : A^\dagger \times A^\dagger \rightarrow (A \times A)^\dagger$  by  $\eta(\Lambda_1, \Lambda_2) = \Lambda_1 \times \Lambda_2$  : m'ble (check)

$D_A := D \cap (A \times A)$ .  $J = \eta^{-1}(\{\Lambda : \#(\Lambda \cap D_A) = 0\})$  : m'ble (check)

$P(\pi_1 \cap \pi_2 \cap A = \emptyset) = 1 \Leftrightarrow (P_1 \times P_2)(J) = 1$ . Use Fubini...

pf of Thm

$N_n(A) = \# \{ \pi_n \cap A \}$ . By Lemma,  $\pi_n$  are disjoint on  $A$ , so  $N(A) = \sum_{n=1}^{\infty} N_n(A)$ .

By countable additivity Thm,  $N(A) \sim \mathcal{P}(\mu)$ ,  $\mu = \mu(A) = \sum_{n=1}^{\infty} \mu_n(A)$

$A_1 \sim A_n$  : disjoint.  $\forall N_n(A_j)$   $n=1, 2, \dots$   $j=1, 2, \dots, k$  are independent

$\Rightarrow N(A_1), N(A_2), \dots, N(A_k)$  are independent.  $\square$

Restriction Thm

$\pi$ : Poisson process on  $S$ , mean measure  $\mu$   $S_i \in S$ .

Then  $\pi_i = \pi \cap S_i$ : Poisson process on  $S$  with mean measure

$$\mu_i(A) = \mu(A \cap S_i).$$

3 The Mapping Thm

Thm  $\pi$ : Poisson process with mean measure  $\mu$ .  $\mu$ :  $\sigma$ -finite  $\left( S = \bigcup_{n=1}^{\infty} S_n, \mu(S_n) < \infty \right)$   
on  $S$ .  $f: S \rightarrow T$  is m'ble f'n,  $\mu^*(B) = \mu(f^{-1}(B))$  is non-atomic.

Then  $f(\pi)$  is a Poisson process on  $T$  with mean measure  $\mu^*$ .

pf) ①  $\mu(S) < \infty$ ,  $A$ : m'ble set.

$\pi_1, \pi_2$ : restrictions of  $\pi$  to  $A$  and  $A^c$ .  $\Rightarrow \pi_1$  &  $\pi_2$  are indep Poisson pro.

$\Rightarrow$  (similar argument to Lemma)  $f(\pi_1)$  &  $f(\pi_2)$  are disjoint with prob 1.

Let  $M$ : measure on  $S \times S$  s.t.  $M(C)$  is # of  $(x, y) \in C$  which  
 $x \in \pi_1, y \in \pi_2$   $f(x) = f(y)$ .

Since  $f(\pi_1)$  &  $f(\pi_2)$  are disjoint,  $M(A \times A^c) = 0$  for  $\forall A$ .

$\Rightarrow \forall A, B \in S, A \times B = \underbrace{[(A \cap B) \times (A \cap B)]}_{\text{in } A \times A^c \text{ or } B \times B^c} \cup \dots$

$$\Rightarrow M(A \times B) = M((A \cap B) \times (A \cap B)) = m(A \cap B) \quad (m(A) := M(A \times A))$$

$$B = S \Rightarrow m(A) = M(A \times S) \Rightarrow m: \text{measure}$$

$M_1$ : induced measure from  $m$  on diagonal  $D \subseteq S \times S$  by  $d: x \mapsto (x, x)$   
i.e.  $M_1(E) := m(d^{-1}(E \cap D)) \Rightarrow M_1(A \times B) = m(A \cap B) = M(A \times B)$

$\Rightarrow$  By uniqueness of product measure,  $M_1 = M$ .

$\Rightarrow M$  is concentrated on  $D$ .  $\Rightarrow \exists x, y \in M$  with  $f(x) = f(y)$  with prob 1.

$$N^*(B) = \# \{f(\pi) \cap B\} = \# \{x \in \pi : f(x) \in B\} = \# \{x \in \pi : x \in f^{-1}(B)\} = N(f^{-1}(B))$$

$$\sim P(\mu(f^{-1}(B))) = P(\mu^*(B))$$

$$\textcircled{2} S = \bigcup_{n=1}^{\infty} S_n, \mu(S_n) < \infty.$$

$\pi_n := \pi \cap S_n \Rightarrow \pi_n$  are independent Poisson process with finite mean measure  $\mu_n (\mu(S_n) < \infty)$

$\Rightarrow f(\pi_n)$  are indep Poisson processes with mean measures

$$\mu_n^*(B) = \mu_n(f^{-1}(B)) \quad \text{by } \textcircled{1}$$

$f(\pi) = f\left(\bigcup_{n=1}^{\infty} \pi_n\right) = \bigcup_{n=1}^{\infty} f(\pi_n)$  is Poisson process with

$$\mu^*(B) = \sum_{n=1}^{\infty} \mu_n^*(B) = \sum_{n=1}^{\infty} \mu(f^{-1}(B) \cap S_n) = \mu(f^{-1}(B))$$

Independence?  $B_1 \sim B_n$ : disjoint  $\Rightarrow f^{-1}(B_1) \sim f^{-1}(B_n)$ : disjoint  $\square$

\* Use of Thm

$\textcircled{1} \pi$ : rate ftn  $\lambda(x_1, x_2, \dots, x_D)$ .

Take  $f: \mathbb{R}^D \rightarrow \mathbb{R}^d$  to be the projection  $f(x_1, \dots, x_D) = (x_1, \dots, x_d)$ ,  $d \leq D$ .

$$\mu^*(B) = \int \dots \int_{B \times \mathbb{R}^{D-d}} \lambda(x_1, \dots, x_D) dx_1 \dots dx_D = \int \dots \int_B \lambda^*(x_1, \dots, x_d) dx_1 \dots dx_d$$

$$\text{where } \lambda^*(x_1, \dots, x_d) = \int_{\mathbb{R}^{D-d}} \lambda(x_1, \dots, x_D) dx_{d+1} \dots dx_D \dots (*)$$

$\mu^*$  is non-atomic

If (\*) converges, then  $f(\pi)$  is a Poisson process with rate  $\lambda^*$  on  $\mathbb{R}^d$ .

$\therefore$  projection of  $\pi$  is again a Poisson process.

$\textcircled{2} \pi$ : uniform with rate  $\lambda$  in the plane.

Transform to plane coordinates by  $f(x, y) = ((x^2 + y^2)^{1/2}, \tan^{-1}(\frac{y}{x}))$

$$\Rightarrow \mu^*(B) = \mu(f^{-1}(B)) = \iint_{f^{-1}(B)} \lambda dx dy = \iint_B \lambda r dr d\theta \Rightarrow \lambda^*(r, \theta) = \lambda r.$$

$\textcircled{3} M(t) = \mu(0, t]$

$M$ : conti, increasing  $\mu(a, b] = M(b) - M(a)$  Take  $f = M$ .

$x = M(t)$ , then  $\mu^*((0, x]) = \mu^*((0, M(t)]) = \mu((0, t]) = M(t) = x$ .

$\Rightarrow U(\Pi)$  is uniform Poisson process of rate 1.

(In ID, the uniform Poisson process is of fundamental importance)

#### [4] Bernoulli process

Recall  $X$ : Bernoulli distribution  $X = \begin{cases} 0 & 1-p \\ 1 & p \end{cases} \quad \mathbb{E}[X] = p$ .

Binomial distribution

$\hookrightarrow$  "n-times"  $\mathbb{E}[X] = np$ .

pick  $A \Rightarrow$  get binomial distribution.

A random finite set  $\Pi \subseteq S$  with  $\# \Pi = n$  and

$$P(N(A_1) = n_1, \dots, N(A_k) = n_k) = \frac{n!}{n_0! n_1! \dots n_k!} p(A_0)^{n_0} p(A_1)^{n_1} \dots p(A_k)^{n_k}$$

is called a Bernoulli process.  $\mathbb{E}[N(A)] = np(A)$

$$P_n(\cdot) = P(\cdot \mid N(S) = n)$$

$$A_0 = S \setminus (A_1 \cup \dots \cup A_k) \\ n_0 = n - \sum_{j=1}^k n_j$$

Let  $A_1, \dots, A_k$  be disjoint subsets of  $S$ . Then

$$\begin{aligned} P_n(N(A_1) = n_1, \dots, N(A_k) = n_k) &= P(N(A_0) = n_0, \dots, N(A_k) = n_k) / P(N(S) = n) \\ &= \frac{\prod_{j=1}^k \frac{e^{-\mu(A_j)} \mu(A_j)^{n_j}}{n_j!}}{\frac{e^{-\mu(S)} \mu(S)^n}{n!}} \\ &= \frac{n!}{n_0! n_1! \dots n_k!} \left( \frac{\mu(A_0)}{\mu(S)} \right)^{n_0} \dots \left( \frac{\mu(A_k)}{\mu(S)} \right)^{n_k} \end{aligned}$$

$\therefore$  Conditioning  $N(S) = n$  converts Poisson to Bernoulli

mean measure  $\mu$  to prob measure  $p(A) = \frac{\mu(A)}{\mu(S)}$

#### [5] The Existence Theorem

Then  $\mu$ : non-atomic measure on  $S$  which can be expressed as  $\mu = \sum_{n=1}^{\infty} \mu_n$ ,  $\mu_n(S) < \infty \Rightarrow \exists$  a Poisson process on  $S$  having  $\mu$  as its mean measure.

pf) WLOG suppose  $\mu_n(S) > 0 \quad \forall n$ .

Construct indep r.v.'s  $N_n, X_{nr} (n, r \in \mathbb{N})$  s.t.

$N_n \sim P(\mu_n(S))$  and  $X_{nr}$  has distribution  $p_n(\cdot) = \frac{\mu_n(\cdot)}{\mu_n(S)}$

Write  $\Pi_n = \{X_{n1}, X_{n2}, \dots, X_{nN_n}\}$ ,  $\Pi = \bigcup_{n=1}^{\infty} \Pi_n$

Write  $\mu_n(A) = \# \{ \Pi_n \cap A \}$ ,

For disjoint  $A_1, A_2, \dots, A_k$  and  $A_0 = S \setminus (A_1 \cup \dots \cup A_k)$

$$P(N_n(A_1) = m_1, \dots, N_n(A_k) = m_k \mid N_n = m) = \frac{m!}{m_0! \dots m_k!} p_n(A_0)^{m_0} p_n(A_1)^{m_1} \dots p_n(A_k)^{m_k}$$

$$m_0 = m - m_1 - \dots - m_k$$

$$\Rightarrow P(N_n(A_1)=m_1, N_n(A_2)=m_2, \dots, N_n(A_k)=m_k)$$

$$= \sum_{m=\sum_{i=1}^k m_i}^{\infty} \frac{e^{-\mu(S)} \mu(S)^m}{m!} \frac{m!}{m_1! m_2! \dots m_k!} \prod_{j=1}^k p_n(A_j)^{m_j}$$

$$= \sum_{m=\sum_{i=1}^k m_i}^{\infty} \pi_{m-\sum_{j=1}^k m_j}(\mu_n(A_0)) \prod_{j=1}^k \pi_{m_j}(\mu_n(A_j)) = \prod_{j=1}^k \pi_{m_j}(\mu_n(A_j))$$

$\Rightarrow N_n(A_j)$  are indep r.v's with distribution  $P(\mu_n(A_j))$

$\Rightarrow \Pi_n$  are indep Poisson processes with mean measure  $\mu_n$

$\therefore$  By Superposition Thm,  $\Pi$  is Poisson process with mean measure  $\mu$